



Course Guide for Basics of Agriculture (HuNu2073)

Program	2 nd Year BSc in Human Nutrition at Bahir Dar Institute of Technology (BiT)				
Module	Agriculture/HuNu-2073				
Course Season	Ac. Year: <u>2023/4</u>		Semester: <u>II</u>		
Course Credits	5 CP				
Student’s Work Load in Hours	Lectures	Tutorials	Lab/Field	Home Study	Total
	3	0	2	3	8
Course Description	The operation of food systems affects diets, health, and nutrition; poverty and food security; energy sources; climate change, environmental sustainability, biodiversity, ecosystems; incomes and employment; and even aspects of peace and governance. There are severe environmental and food injustices. The course is designed as a support course for students to knowledge, skills and attitude of nature-positive agriculture in aiding them support the agriculture sector for improved nutrition as well as resilient, sustainable, sovereign and accessible food for all.				
Course Objectives and Competences to be Acquired	After completing this course, students will be able to: <i>Describe climate, climate change, natural resource (NR), and its management</i> Describe the basic principles of plant and animal production Discuss the production and management of crops Define key terminology related to animal science and production Identify the <i>implication of NR /e.g. soil/ or agri. production on nutrition and health</i>				
Course Contents	<i>The role of nature/environment (soil, water, forest and air/atmosphere)</i> <i>Soil erosion, conservation and its relevance for plant (and animal production)</i> Production of different crops and improved cropland management The basics of efficient animal production The basic fact on feeds and nutrient of animals				
Pre-requisites	None				
Course Status	Compulsory				
Week 1	Natural Resources Management 1.1. Introduction to natural resources 1.2. Soil 1.2.1. Definition /http: <u>CSA News - 2017 - Es - A New Definition of Soil/</u> 1pp. 1.2.2. Formation 1.3. Soils and Plants 1.3.1. Essential Plant Nutrients				
Week 2	1.3.2. Soil-Plant Relations				

	1.4. Key Pillars of Natural Resource Management 1.5. Natural Resources (NR) Degradation and Conservation 1.5.1. Natural Resources (NR) Degradation 1.5.1.1. Soil Degradation 1.5.1.2. Causes of Natural Resource (NR) Degradation 1.5.2. Environment/ Natural Resource Conservation 1.5.2.1. Soil and water conservation
References on the above topics at Jan-Moscow Library (https://catalog.bdu.edu.et)	
1. Beasley, R.P., J.M. Gregory and T.R. McCarty, 1984. Erosion and sediment pollution control, 2nd ed. Iowa State University Press, Ames, Iowa. Chapter 7. Terraces. pp. 154-165. Library code: 631.45BEA on Shelf <u>064C</u> 2. Cunningham, W.P., and Saigo, B.W., 1995. Environmental Science: a global concern, 3rd ed. Wm. C. Brown Publishers, London. Chapter 11. Soil Resources and Sustainable Agriculture. pp. 218-236. Library code: 363.7/8CUN on Shelf <u>095C</u> 3. Dasmann, R.F., 1984. Environmental Conservation, 5 th ed. John Wiley & Sons, Inc., New York, US. 486p. pp. 11-16, pp. 180-181. Library code: 333.7:2 on Shelf <u>087B</u> 4. Delroit, R.J., 1967 (Old Reference). Crop production, 3rd ed. Prentice Hall, New Jersey, US. pp. 599-613 (Crop Rotaion and Green Manuring). Library code: 631.5DEL on Shelf <u>064C</u> 5. Dupriez, H., 1992. Land & Life: Ways of Water Run-Off, Irrigation and Drainage. Philippe De Leener, NEW DELHI. pp.161-163. Library code: 631.587DUP on Shelf <u>064C</u> 6. Fitzpatrick, E.A., 1986. An Introduction to Soil Science. pp.10-64. Factors of Soil Formation & Processes in the Soil System. Library code: 631.4FIT on Shelf <u>064C</u> 7. Hudson, N., 1971(Old Reference). Soil Conservation. B T Batsford Limited, London. pp. 85-95; pp. 201-210 (11.3&11.4) Library code: 631.45HUD on Shelf <u>064C</u> 8. Huffman, R. L., D. D. Fangmeier, W. J. Elliot, S. R. Workman, and. G. O. Schwab, 2011. Soil and Water Conservation Engineering. 6 th ed. ASABE, US. 523p. Library code: 631.45SCH on Shelf 008 Huffman et al., 2013 7 th ed. Online at book download sites such as http://libgen.rs . 9. Lockhart, J.A.R., and A.J.L. Wiseman, 1970. Introduction to crop husbandry, 2nd ed. Pergamon Press, Oxford, G. Britain. pp. 35-39 and pp. 209. Climate and weather Library code: 631.5LOC on Shelf <u>064C</u> 10. Ministry of agriculture natural resource conservation & dev't, 1986. Guidelines for Development Agents on Soil Conservation in Ethiopia. Community Forests & Soil Cons. Dev't Dept (CFSCDD). pp. 37-78 Library code: 631.4963MIN on Shelf <u>064C</u> <u>NEW/ONLINE EDITION</u> Hurni, H., Berhe, W.A., Chadhokar, P., Daniel, D., Gete, Z., Grunder, M., Kassaye, G., 2016. Soil and Water Conservation in Ethiopia: Guidelines for Development Agents, 2nd revised edition. Centre for Development and Environment (CDE), University of Bern, with Bern Open Publishing (BOP), Bern, Switzerland. pp. 53-116. <u>Online@</u> https://boris.unibe.ch/80013/1/Guidelines_Soil_and_Water_Conservation_in_Ethiopia	

2016.pdf	
11. O’Riordan, T., 1995. Environmental Science for Environmental Management. Longman, Essex, England. 369p . Chapter 5. Ecosystem management by Barkham, J., pp. 83-93; Chapter 12. Soil erosion and land degradation by Stocking, M., pp. 225-241. Library code: 363.7ENV OR 363.7O’RI OR 363.7ORI on Shelf <u>095C</u>	
12. Tolanur, S., 2004. Fundamentals of soil science. pp.1-64. International Book Distributing Co. (Publishing Division), Lucknow, India. Library code: 631.4TOL on Shelf <u>064C</u>	
13. Suresh, R., 2012. Soil and Water Conservation Engineering. Library code: 333.7316sur <u>Also Online at book download sites</u> such as http://libgen.rs ; https://www.pdfdrive.com .	
14. Tripathi, R.P. and H.P. Singh, 1993. Soil Erosion and Conservation. New International (P) Limited, Publishers, New Delhi. 305p. Library code: 631.451TRI on Shelf <u>064C</u>	
15. Young, A., 1989. Agroforestry for Soil Conservation. CABI, Exeter, UK. Chapter 2-3. Soil Conservation and Sustainability; Agroforestry. pp. 9-13: Code: 631.4913YOU on Shelf <u>064C</u> .	
Week 3	1.5.2.2. Impact of NR conservation to food, post-harvest and nutrition 1.5.3. Food biodiversity and nutrition
Weeks 4-7	Crop Production and Management Introduction to crop production Basic science of agronomic practices of crop production Preparation of the soil Sowing methods and Spacing Fertilizer application Irrigation Protection measures Harvesting Production and management of major crops in Ethiopia: Fruits, Vegetables, Cereals, legumes, oilseeds and tubers
Weeks 8 -10	Animal Production Introduction Livestock and small ruminant husbandry Basic science of apiculture, poultry and fish production and utilization Animal feeding and health Animal production linkage to food/role of animal production in food production
Week 11  Week 12	Climatic Change in Agriculture and Actions What is climate, weather and climate change Factors affecting climate change 4.1. Introduction to Climate Change 4.2. Climate Change and Agri-food System 4.2.1. Effects of climate change on agriculture 4.2.2. Impacts of climate on food and nutrition security 4.2.3. Effects of agriculture in climate change 4.3. Climate-resilient interventions 4.3.1. Mitigation
References on the above topics at Jan-Moscow Library (https://catalog.bdu.edu.et)	

16. Hadlow, L., 1960 (Very old). Climate, Vegetation & Man, 4 th ed. Fundamental Geography III. London. pp. 51. <i>Climate and weather</i> . Library code: 910HAD on Shelf <u>112E</u>		
17. Masters, G.M., 1991. Introduction to Environmental Engineering and Science. Prentice Hall, New Jersey, US. pp. 391 (8.4. The C Cycle); pp. 421-422 (8.6. Other Greenhouse Gases) Library code: 628 MAS on Shelf <u>063B</u>		
18. O’Riordan, T., 1995. Environmental Science for Environmental Management. Longman, Essex, England. Chapter 6. The threat of global warming by Clayton, K., pp. 113-118. Library code: 363.7ENV OR 363.7O’RI OR 363.7ORI on Shelf <u>095C</u>		
19. Porteous, A., 1991. Dictionary of Environmental Science and Technology. Libr. Code: 600 POR		
20. Wittwer, S.H., 1995. Food, Climate, and Carbon Dioxide: The Global Environment and World Food Production. 236p . Library code: 338.14WIT on Shelf <u>094E</u>		
Weeks 13-14	Food value chain Definition of value chain in the food system The advantages of food value chain Food safety and quality along food value chain The role of different actors in food value chain Value addition in food value chain Resilient value chain in food system	
Weeks 15-16	Final Exam	
t-Learning Methods	Interactive Lectures, small group activities, brainstorming, field demonstration and reports, reading, and assignment	
Practical field visit ➤ different crops/plant foods (including vegetables, fruits, and tubers) and their storage ➤ different animal production (and product processing)		
Assessment Method	Based on the criteria of BiT or Bahir Dar University	Weight
Individual Reports: crops (10%) and animal production [and product processing] (10%)		20%
Mid-term (30%) and Final (50%) Examination		80%

Recent References Found Only Online

FAO, 2022. Soils for Nutrition: State of the Art. FAO, Rome. <https://doi.org/10.4060/cc0900en>

Owino V, Kumwenda C, Ekesa B, Parker ME, Ewoldt L, Roos N, Lee WT and Tome D (2022) The impact of climate change on food systems, diet quality, nutrition, and health outcomes: A narrative review. Front. Clim. 4:941842. doi: 10.3389/fclim.2022.941842
<https://www.frontiersin.org/articles/10.3389/fclim.2022.941842/full>

Weil, R.R., and N.C. Brady, 2017. The Nature and Properties of Soils, 15th ed, Pearson Education, Essex, England. Online at book download sites such as <http://libgen.rs>; <https://www.pdfdrive.com>.

[AR6 Synthesis Report: Climate Change 2023 — IPCC](#) Summary for Policymakers OR Longer Report
<https://climatepromise.undp.org/news-and-stories/climate-dictionary-everyday-guide-climate-change>

https://www.ipcc.ch/site/assets/uploads/2022/02/PR_WGII_AR6_english.pdf

Chitando, E., E.M. Conradie, S.M. Kilonzo (*Eds.*), 2022. African Perspectives on Religion and Climate Change. Routledge, an imprint of the Taylor & Francis Group, an informa business. Introduction: Ubuntu and Indigenous Knowledge <https://doi.org/10.4324/9781003147909>